1 Chapter 1

All:1.1,1.2,1.7,1.14 and 6 examples

1.1:

$$x = tan\theta d \quad \dot{x} = \frac{\dot{\theta}}{cos^2\theta}d = \frac{\omega}{cos^2\theta}d$$
$$cos\theta = \frac{d}{\sqrt{d^2+x^2}} \Rightarrow \frac{\omega(d^2+x^2)}{d}$$
$$\ddot{x} = \frac{2x\dot{x}\omega}{d} = \frac{2x\omega}{d}\frac{\omega(d^2+x^2)}{d} = \frac{2x\omega^2(d^2+x^2)}{d^2}$$

1.2:

$$\begin{cases} x = sin\theta b \\ y = cos\theta d \end{cases}$$
$$\frac{d}{dt}cos\theta(b+d) = -c$$
$$\Rightarrow \dot{\theta} = \frac{c}{sin\theta(b+d)}$$
$$\begin{cases} \dot{x} = \frac{cbcos\theta}{sin\theta(b+d)} \\ \dot{y} = -\frac{dc}{(b+d)} \end{cases}$$
$$\begin{cases} \ddot{x} = -\frac{c^2b}{sin^3\theta(b+d)^2} \\ \ddot{y} = 0 \end{cases}$$

1.7:

$$\vec{v} = v_x\vec{i} + v_y\vec{j} \quad v_x^2 + v_y^2 = c$$
$$a = \dot{v_x}\vec{i} + \dot{v_y}\vec{j}$$
$$\vec{a}\cdot\vec{v} = v_x\dot{v_x} + v_y\dot{v_y} = \frac{1}{2}(\dot{v_x^2} + \dot{v_y^2}) = \frac{1}{2}\dot{c} = 0$$

1.14:

$$\begin{cases} m\ddot{x} = -mk\dot{x} \\ m\ddot{y} = -mk\dot{y} - mg \end{cases}$$

$$\begin{cases} \dot{x} = C_1 e^{-kt} \\ \dot{y} = C_2 e^{-kt} - \frac{g}{k} \end{cases}$$

$$t = 0 \quad C_1 = v_{x0}; C_2 = \frac{g}{k} + v_{y0}$$

$$-tan\alpha = \frac{(\frac{g}{k} + v_{y0})e^{-kt} - \frac{g}{k}}{v_{x0}e^{-kt}}$$

$$-tan\alpha = (\frac{g}{kv_{x0}} + tan\alpha) - \frac{g}{kv_{x0}}e^{kt}$$

$$\Rightarrow e^{kt} = \frac{kv_{x0}}{g}(\frac{g}{kv_{x0}} + 2tan\alpha)$$

$$t = \frac{1}{k}ln[\frac{kv_{x0}}{g}(\frac{g}{kv_{x0}} + 2tan\alpha)] = \frac{1}{k}ln[1 + 2\frac{k}{g}sin\alpha v_0]$$

exp1:

$$v = \begin{cases} v_x = v_0 \\ v_y = gt \end{cases}$$

$$a_n = \frac{v^2}{\rho} \Rightarrow \rho = \frac{v^2}{a_n}$$

$$a_n = \vec{a} \cdot \vec{n} = \sqrt{a^2 - (a \cdot v\frac{1}{|v|})^2}$$

$$\vec{a} = g\vec{j}; \vec{a} \cdot \vec{v} = g^2 t$$

$$a_n = \sqrt{g^2 - \frac{g^4t^2}{v_0^2 + g^2t^2}} = \sqrt{\frac{v_0^2 g^2}{v_0^2 + g^2t^2}} = \frac{v_0 g}{\sqrt{v_0^2 + g^2t^2}}$$

$$\rho = \frac{(v_0^2 + g^2t^2)^{\frac{3}{2}}}{v_0 g}$$

exp2:

$$m\ddot{y} = -mg - mk\dot{y}$$

$$\ddot{y} + k\dot{y} = -g \Rightarrow y = A + Be^{-kt} - \frac{gt}{k}$$

$$\dot{y}|_{t=0} = 0 \Rightarrow \dot{y} = \frac{g}{k}e^{-kt} - \frac{g}{k}$$

exp3:

$$l = r \times p = 2 \times 0.5 \times 4 = 1 = v_B \times 0.5 \times 2$$
$$\Rightarrow v_B = 1$$

exp4:

$$mg = kx \Rightarrow x = \frac{mg}{k}$$
$$\frac{1}{2}mv^2 - mgx + \frac{1}{2}kx^2 = 0$$
$$\Rightarrow \frac{1}{2}mv^2 = \frac{m^2g^2}{k} - \frac{1}{2}k\frac{m^2g^2}{k^2} = \frac{m^2g^2}{2k}$$
$$v = \sqrt{\frac{m}{k}}g$$

exp5:

$$V = -\int (ax + by^2)dx + (az + 2bxy)dy + (ay + bz^2)dz$$
$$= -\frac{1}{2}ax^2 - \frac{1}{3}bz^3 - \int by^2dx + azdy + bxdy^2 + aydz$$
$$= -\frac{1}{2}ax^2 - \frac{1}{3}bz^3 - by^2x - azy + C$$
$$C = 0$$

exp6:

$$\frac{d^2x}{dt^2} = 2 + 6x^2$$
$$\int_0^x adx = 2x + 2x^3 = \int_0^x \frac{d^2x}{dt^2}dx = \int_0^x vdv = \frac{1}{2}v^2 - \frac{1}{2}v_0^2$$
$$v = \sqrt{100 + 4x + 4x^3}$$

2 Chapter 2

2.8,2.13,2 examples

2.8:

$$\vec{r} = r\vec{e}_r; \vec{v} = \dot{r}\vec{e}_r + r\omega\vec{e}_\theta;$$

$$\vec{a} = \ddot{r}\vec{e}_r + \dot{r}\omega\vec{e}_\theta + (\dot{r}\omega + r\dot{\omega})\vec{e}_\theta - r\omega^2\vec{e}_r = (\ddot{r} - r\omega^2)\vec{e}_r + (2\dot{r}\omega + r\dot{\omega})\vec{e}_\theta$$

$$r \times p = h \Rightarrow r^2\omega = \frac{h}{m}$$

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta}\omega = \frac{dr}{d\theta}\frac{h}{mr^2} = -\frac{h}{m}\frac{d\frac{1}{r}}{d\theta}$$

$$r = ae^{b\theta} \Rightarrow \dot{r} = -\frac{h}{m}\frac{1}{a}(-b)e^{-b\theta} = \frac{bh}{m}\frac{1}{r}$$

$$\ddot{r} = -\frac{hb}{m}\frac{1}{r^2}(\frac{hb}{m}\frac{1}{r}) = -\frac{h^2b^2}{m^2}\frac{1}{r^3}$$

$$\ddot{r} - r\omega^2 = \frac{F}{m} = -\frac{k}{mr^3} = -\frac{h^2b^2}{m^2}\frac{1}{r^3} - r\frac{h^2}{m^2r^4}$$

$$\Rightarrow k = \frac{h^2b^2}{m} + \frac{h^2}{m} \Rightarrow h = \sqrt{\frac{km}{b^2+1}}$$

$$E = T + V = \frac{1}{2}m(\dot{r}^2 + r^2\omega^2) - \frac{k}{2r^2} = \frac{1}{2}m(\frac{h^2b^2}{m^2r^2} + \frac{h^2}{m^2r^2}) - \frac{k}{2r^2}$$

$$\Rightarrow \frac{1}{2r^2}(\frac{h^2b^2}{m} + \frac{h^2}{m} - k) = 0$$

2.13:

$$V = \frac{mc}{2r^2}$$

$$\frac{1}{2}mv_\infty^2 = \frac{1}{2}mv^2 + \frac{mc}{2d^2}$$

$$\rho v_\infty = vd \Rightarrow v = \frac{\rho v_\infty}{d}$$

$$= v_\infty^2 - \frac{\rho^2 v_\infty^2}{d^2} = \frac{c}{d^2}$$

$$d = \sqrt{\frac{c + \rho^2 v_\infty^2}{v_\infty^2}}$$

exp1:

$$\begin{aligned}
&3av_0 = av \Rightarrow v = 3v_0\\
&\frac{2k}{3a} = 4mv_0^2 \Rightarrow k = 6amv_0^2\\
&\frac{1}{2}mv_0^2 + \frac{k}{6a} = \frac{1}{2}mv^2\\
&\Rightarrow \frac{3}{2}mv_0^2 = \frac{1}{2}mv^2\\
&v = \sqrt{3}v_0
\end{aligned}$$

exp2:

$$\begin{aligned}
&l = cos\alpha 2Rmv_0 = mvR\\
&v = 2cos\alpha v_0\\
&\frac{mv_0^2}{2R} = \frac{k}{4R^2} \Rightarrow k = 2Rmv_0^2\\
&\frac{3}{2}mv_0^2 = \frac{1}{2}mv^2\\
&v = \sqrt{3}v_0\\
&cos\alpha = \frac{\sqrt{3}}{2} \Rightarrow \alpha = \frac{\pi}{6}
\end{aligned}$$

3 Chapter 3

2 expamples

exp1:

$$\begin{aligned}
&r = r_c + r'; v = \dot{r}' + \omega \times r'; a = \ddot{r}' + \omega \times \dot{r}' + \omega \times \omega \times r' + \omega \times \dot{r}'\\
&r = r' = b\vec{i} + (b - ut)\vec{j}\\
&\vec{v} = -u\vec{j} + \omega((b - ut)\vec{i} - b\vec{j})\\
&\vec{a} = 0 - 2\omega u\vec{i} - 2\omega^2(b - ut)\vec{j} - 2b\omega^2\vec{i} + \omega \times \omega(-b\vec{j} + (b - ut)\vec{i})\\
&= -2\omega u\vec{i} - 2\omega^2(b - ut)\vec{j} - 2b\omega^2\vec{i} - \omega^2 b\vec{i} - \omega^2(b - ut)\vec{j}\\
&|v| = \sqrt{\omega^2(b - ut)^2 + (\omega b + u)^2}\\
&|a| = \sqrt{(-\omega^2 b - 2b\omega^2 - 2\omega u)^2 + (-2\omega^2(b - ut) - \omega^2(b - ut))^2}
\end{aligned}$$

3.5:

$$m\vec{a}' = -mg\vec{j} - ma\vec{i} + \vec{T}$$
$$\begin{cases} ma_1 = sin\alpha mg - ma_0 cos\alpha \\ T = cos\alpha mg + sin\alpha ma_0 \end{cases}$$

4 Chapter 4

4.1 eg.4.2 and 4 examples

4.1:

$$x = rsin\theta; y = r\cos\theta$$
$$\begin{cases} \bar{x} = \frac{\int xdm}{\int dm} = 0 \\ \bar{y} = \frac{\int ydm}{\int dm} \end{cases}$$
$$\bar{y} = \frac{2r^2 sin\frac{s}{2r}}{s}$$

eg.4.2:

$$L = L_c + L'$$
$$\omega r = \Omega(R - r)$$
$$L_c = \Omega(R - r)^2 3m\vec{e}_1$$
$$L' = -3mr^2\omega\vec{e}_1$$
$$L = 3m\omega(rR - 2r^2)$$
$$T = T_c + T'$$
$$T_c = \frac{1}{2}3m\omega^2 r^2$$
$$T' = \frac{1}{2}3m\omega^2 r^2$$
$$T = 3m\omega^2 r^2$$

exp1:

$$T = T_c + T'$$
$$T_c = \frac{1}{2}4mv_c^2$$
$$T' = \frac{1}{2}4mr^2\omega s^2$$
$$\omega r = v_c$$
$$T = 4mv_c^2$$

exp2:

$$
\begin{aligned}
&mv_0 = (M+m)v \Rightarrow v = \frac{m}{M+n}v_0 \\
&\frac{1}{2}mv_0^2 = \frac{1}{2}(M+m)v^2 + \mu mgs \\
&\Rightarrow \frac{1}{2}mv_0^2 - \frac{1}{2}\frac{m^2}{M+m}v_0^2 = \mu mgs \\
&\Rightarrow \frac{M}{2\mu g(M+m)}v_0^2 = s
\end{aligned}
$$

exp3:

$$
\begin{aligned}
&mv_0 = mv_1 + mv_2 \Rightarrow v_2 = v_0 - v_1 \\
&\frac{1}{2}mv_0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 \\
&v_0v_1 = v_1^2 \Rightarrow v_1 = \begin{cases} v_0 \\ 0 \end{cases} \\
&v_2 = \begin{cases} 0 \\ v_0 \end{cases} \qquad v_2 \neq 0 \\
&v_1 = 0
\end{aligned}
$$

exp4:

$$
\begin{aligned}
&l = r \times p = lmv_0 \\
&lmv_0 = L' + L_c \\
&mv_0 = 3mv \Rightarrow v = \frac{1}{3}v_0 \\
&L_c = \frac{2}{3}l3m\frac{1}{3}v_0 = \frac{2}{3}mlv_0 \\
&L' = \frac{1}{3}l2m\omega\frac{1}{3}l + \frac{2}{3}lm\omega\frac{2}{3}l = \frac{2}{3}ml^2\omega \\
&\Rightarrow lmv_0 = \frac{2}{3}mlv_0 + \frac{2}{3}ml^2\omega \\
&\Rightarrow \omega = \frac{v_0}{2l}
\end{aligned}
$$

5 Chapter 5

5.8,5.10,5.11,5.17,5.21,5.22,eg.5.4 and 8 examples.

5.8:

$$L^a = J^a_b \omega^b$$

$$J_{ij} = \begin{bmatrix} \int y^2 + z^2 dm & -\int xydm & -\int zxdm \\ -\int yxdm & \int x^2 + z^2 dm & -\int yzdm \\ -\int zxdm & -\int yzdm & \int x^2 + y^2 dm \end{bmatrix}$$

$$dm = \frac{m}{l} dz; x = y = 0$$

$$J_{ij} = \begin{bmatrix} \frac{1}{3} ml^2 & 0 & 0 \\ 0 & \frac{1}{3} ml^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\vec{\omega} =] [$$

$$\vec{L} = \frac{1}{3} ml^2 sin\theta \omega \vec{i}$$

5.10:

$$df = \mu g dm \quad dm = \frac{m}{\pi R^2} ds = \frac{m}{\pi R^2} r dr d\theta$$

$$M = \int r \times df = \int \frac{m}{\pi R^2} r^2 dr d\theta = 2\pi \frac{m}{\pi R^2} \frac{1}{3} R^3 = \frac{2}{3} m \mu g R$$

$$M \Delta t = J \Delta \omega$$

$$\Rightarrow \frac{2}{3} \mu m g R \Delta t = \frac{1}{2} m R^2 \omega_0$$

$$\Delta t = \frac{3}{4} \frac{\omega_0 R}{\mu g}$$

5.11:

$$
\begin{aligned}
&df = kv^2 ds = k\omega^2 x^2 l dx \\
&M = \int r \times df = \int k\omega^2 x^3 l dx = \frac{1}{4} k\omega^2 d^4 l \\
&M = J\beta \\
&J = \int x^2 dm = \int x^2 \frac{m}{d} dx = \frac{1}{3} m d^2 \\
&\frac{d\omega}{dt} = -\frac{3}{4m} k\omega^2 l d^2 \\
&\Rightarrow -\frac{d\frac{1}{\omega}}{dt} = -\frac{3}{4m} k l d^2 \\
&\frac{1}{\omega_0} - \frac{2}{\omega_0} = -\frac{3}{4m} k l d^2 t \\
&t = \frac{4m}{3kld^2\omega_0}
\end{aligned}
$$

5.17:

$$
\begin{aligned}
&(i) omitted \\
&(ii) h = \frac{1}{2} g t^2 \\
&\frac{1}{3} m l^2 \omega^2 = mgl \Rightarrow \omega = \sqrt{\frac{3g}{l}} \\
&\Delta\theta = \omega \Delta t = \sqrt{\frac{3g}{l}} \sqrt{\frac{2h}{g}} = \sqrt{\frac{6h}{l}} \\
&round = \frac{1}{2\pi} rad = \frac{1}{\pi} \sqrt{\frac{3h}{2l}}
\end{aligned}
$$

5.21:

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\frac{2}{5}mR^2\dot{\theta}^2 + \int \mu mg dx - \int \mu mgR d\theta$$

$$\frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{q}}) = \frac{\partial \mathcal{L}}{\partial q}$$

$$\begin{cases} m\ddot{x} = \mu mg \\ m\frac{2}{5}R^2\ddot{\theta} = -\mu mgR \end{cases}$$

$$\Rightarrow \begin{cases} \dot{x} = \mu gt \\ \dot{\theta} = \omega_0 - \frac{5}{2}\mu\frac{g}{R}t \end{cases}$$

$$\dot{x}|_t = \omega|_t R$$

$$\mu gt = \omega_0 R - \frac{5}{2}\mu gt$$

$$t = \frac{2}{7}\frac{\omega_0 R}{\mu g}$$

$$x = \frac{1}{2}\mu gt^2 = \frac{1}{2}\frac{4}{49}\frac{\omega_0^2 R^2}{\mu g} = \frac{2}{49}\frac{\omega_0^2 R^2}{\mu g}$$

5.22:

$$F cos\theta R = M$$

$$cos\theta = \frac{h-R}{R}$$

$$M\Delta t = \frac{2}{5}mR^2\omega_0 = FR\frac{h-R}{R}\Delta t$$

$$F\Delta t = mv_0 \Rightarrow \Delta t = \frac{mv_0}{F}$$

$$v_0 = \omega_0 R$$

$$(h-R) = \frac{2}{5}R$$

$$h = \frac{7}{5}R$$

eg.5.4:

$$J_{ij} = \begin{bmatrix} \int y^2 + z^2 dm & -\int xydm & -\int zxdm \\ -\int yxdm & \int x^2 + z^2 dm & -\int yzdm \\ -\int zxdm & -\int yzdm & \int x^2 + y^2 dm \end{bmatrix}$$
$$dm = \frac{m}{a^3} dxdydz$$
$$J_{ij} = \begin{bmatrix} \frac{2}{3}ma^2 & -\frac{1}{4}ma^2 & -\frac{1}{4}ma^2 \\ -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 & -\frac{1}{4}ma^2 \\ -\frac{1}{4}ma^2 & -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 \end{bmatrix}$$
$$J = e_{\omega a} J^a_b e^b_\omega$$
$$e_\omega = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$
$$J^a_b e^b = \frac{1}{\sqrt{3}} \begin{bmatrix} \frac{1}{6}ma^2 \\ \frac{1}{6}ma^2 \\ \frac{1}{6}ma^2 \end{bmatrix}$$
$$e_a J^a_b e^b = \frac{1}{3}(\frac{1}{2}ma^2) = \frac{1}{6}ma^2$$
$$T = \frac{1}{2}\omega_a J^a_b \omega^b = \frac{1}{2}\omega^2 J = \frac{1}{12}ma^2\omega^2$$

exp1:

$$\mathcal{L} = \frac{1}{2}m\dot{l}^2 + \frac{1}{2}\frac{1}{2}mR^2\omega^2 - (c - mgsin\alpha l) - fl + \int fRd\theta$$
$$\begin{cases} m\ddot{l} = mgsin\alpha - f \\ \frac{1}{2}mR^2\frac{d\omega}{dt} = fR \end{cases}$$
$$\ddot{l} = \dot{\omega}R$$
$$mgsin\alpha - f = 2f$$
$$\frac{1}{3}mgsin\alpha = f$$

exp2:

$$J_{ij} = \begin{bmatrix} \int y^2+z^2dm & -\int xydm & -\int zxdm \\ -\int yxdm & \int x^2+z^2dm & -\int yzdm \\ -\int zxdm & -\int yzdm & \int x^2+y^2dm \end{bmatrix}$$

$$y = rsin\theta; x = rsin\theta, z = 0; dm = \frac{m}{\pi R^2}rdrd\theta$$

$$J_{ij} = \begin{bmatrix} \frac{1}{4}mR^2 & 0 & 0 \\ 0 & \frac{1}{4}mR^2 & 0 \\ 0 & 0 & \frac{1}{2}mR^2 \end{bmatrix}$$

$$\vec{\omega} = \begin{bmatrix} \frac{\sqrt{3}}{2}\omega \\ 0 \\ \frac{1}{2}\omega \end{bmatrix}$$

$$T = \frac{1}{2}\omega J\omega = \frac{1}{2}\omega^2 \begin{bmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{\sqrt{3}}{8}mR^2 \\ 0 \\ \frac{1}{4}mR^2 \end{bmatrix} = \frac{1}{2}\omega^2(\frac{3}{16}mR^2 + \frac{1}{8}mR^2)$$

$$= \frac{5}{32}\omega^2 mR^2$$

exp3:

$$\frac{1}{2}M\omega^2 l^2 + \frac{1}{2}\frac{1}{3}ml^2\omega^2 - Mgl - mg\frac{1}{2}l = 0$$

$$(2M+m)gl = (Ml^2 + \frac{1}{3}ml^2)\omega^2$$

$$\omega = \sqrt{\frac{(2M+m)g}{(M+\frac{1}{3}m)l}}$$

exp4:

$$y = \frac{1}{2}lsin\theta \Rightarrow v = \frac{1}{2}lcos\theta\dot{\theta}$$

$$\frac{1}{2}m\dot{y}^2 - mg(1-sin\theta)\frac{1}{2}l = 0$$

exp5:

$$\mathcal{L} = \frac{1}{2}m_1\dot{l}^2 + \frac{1}{2}m_2\dot{l}^2 + \frac{1}{2}\frac{1}{2}m\dot{l}^2 - (C - m_1gl) - \int \mu m_2 g dl$$
$$\ddot{l} = \frac{(m_1 - \mu m_2)g}{(m_1 + m_2 + \frac{1}{2}m)}$$
$$T_1 - m_1g = -m_1\ddot{l} \Rightarrow T_1 = m_1(-\frac{(m_1 - \mu m_2)g}{(m_1 + m_2 + \frac{1}{2}m)} + 1)$$
$$T_1 = m_1(\frac{(m_2 + \mu m_2 + \frac{1}{2}m)g}{m_1 + m_2 + \frac{1}{2}m})$$
$$m_2\ddot{l} = T_2 - \mu m_2 g \Rightarrow T_2 = m_2(\ddot{l} + \mu g)$$
$$T_2 = m_2(\frac{(1+\mu)m_1 + \frac{1}{2}\mu m}{m_1 + m_2 + \frac{1}{2}m})g$$

exp6:

$$T = \frac{1}{2}mv^2 + \frac{1}{2}\frac{2}{5}mR^2\omega^2$$
$$\frac{mv^2}{2R} = mgcos\theta$$
$$v = \omega R$$
$$\frac{1}{2}mv^2 + \frac{1}{5}mv^2 = mg(1 - cos\theta)2R$$
$$\frac{7}{10}mv^2/2R = mg(1 - cos\theta)$$
$$\frac{7}{10}cos\theta = 1 - cos\theta$$
$$cos\theta = \frac{10}{17} \Rightarrow \theta = arccos(\frac{10}{17})$$

exp7:

$$
\begin{aligned}
&l = \frac{1}{3}ml^2\omega_0 \\
&\frac{1}{2}\frac{1}{3}ml^2\omega_0^2 = mg\frac{1}{2}l \Rightarrow \omega = \sqrt{\frac{3g}{l}} \\
&\frac{1}{3}ml^2\omega_0 = \frac{4}{3}ml^2\omega \Rightarrow \omega = \frac{1}{4}\omega_0 \\
&(1 - cos\theta) = \frac{1}{12} \\
&cos\theta = \frac{11}{12} \\
&\theta = arccos(\frac{11}{12})
\end{aligned}
$$

exp8:

$$
\begin{aligned}
&m\omega^2\frac{1}{2}l = -\frac{1}{2}mg + F_1 \\
&l^2\omega^2 = g\frac{3}{2}l \Rightarrow \omega = \sqrt{\frac{3g}{2l}} \\
&F_1 = \frac{5}{4}mg \\
&M\beta = mgl\frac{\sqrt{3}}{4} = \frac{1}{3}ml^2\beta \Rightarrow \beta = \frac{3\sqrt{3}}{4}\frac{g}{l} \\
&a_1 = \beta\frac{1}{2}l = \frac{3\sqrt{3}}{8}g \\
&\frac{3\sqrt{3}}{8}gm = \frac{\sqrt{3}}{2}gm - F_1 \Rightarrow F_1 = \frac{\sqrt{3}}{8}mg \\
&|F| = \sqrt{\frac{3}{64} + \frac{25}{16}}mg = \frac{\sqrt{103}}{8}mg
\end{aligned}
$$

6 Chapter 6

6.1 6.7 6.10 and 6 examples.

6.1:

$$q = \theta$$
$$\frac{h-y}{h} = \frac{\frac{1}{2}l_0}{l}$$
$$h = sin\theta l$$
$$\Rightarrow sin\theta l - y = \frac{1}{2}l_0 sin\theta \Rightarrow y = sin\theta l - \frac{1}{2}l_0 sin\theta$$
$$2cos\theta r = l \Rightarrow y = sin(2\theta)r - \frac{1}{2}sin\theta l_0$$
$$\delta y = 2cos2\theta r - \frac{1}{2}cos\theta l_0; \delta w = mg\delta y = 0$$
$$4cos2\theta r - cos\theta l_0 = 0$$
$$0 = 8cos^2\theta r - 4r - cos\theta l_0 \Rightarrow cos\theta = \frac{l_0 \pm \sqrt{l_0^2 + 4(8r \times 4r)}}{16r}$$
$$cos\theta > 0 \Rightarrow cos\theta = \frac{l_0 + \sqrt{l_0^2 + 128r^2}}{16r}$$
$$l = 2cos\theta r$$
$$\Rightarrow l = \frac{l_0 + \sqrt{l_0^2 + 128r^2}}{8}$$
$$\Rightarrow 8l - l_0 = \sqrt{l_0^2 + 128r^2} \Rightarrow 64l^2 - 16ll_0 = 128r^2$$
$$l_0 = \frac{64l^2 - 128r^2}{16l} = \frac{4l^2 - 8r^2}{l}$$

exp1:

$$\vec{r}_A = sin\phi l\vec{i} - cos\phi l\vec{j}; F_A = F\vec{i}$$
$$\vec{r}_C = \frac{1}{2}sin\phi l\vec{i} - \frac{1}{2}cos\phi l\vec{j}; F_C = -mg\vec{j}$$
$$\delta w = (cos\phi\delta\phi l\vec{i} + sin\phi\delta\phi l\vec{j})F\vec{i} + (\frac{1}{2}sin\phi\delta\phi l\vec{j})(-mg\vec{j})$$
$$0 = cos\phi F - mg\frac{1}{2}sin\phi$$
$$\phi = arctan\frac{2F}{mg}$$

exp2:

$$
\begin{aligned}
&q = \alpha \\
&L = -cot\alpha a + 2cos\alpha l \\
&y = -L\vec{j} \\
&\delta y = -\frac{1}{sin^2\alpha}a + 2sin\alpha l = 0 \\
&\alpha = arcsin(\frac{a}{2l})^{\frac{1}{3}}
\end{aligned}
$$

exp3:

$$
\begin{aligned}
&r_a = sin\theta lj; F_A = -mgj \\
&r_B = cos\theta i; F_B = -Fi \\
&\delta w = 0 = -mgjcos\theta lj\delta\theta + sin\theta\delta\theta F \\
&F = cot\theta mg
\end{aligned}
$$

6.7:With Lagrangian

$$
\begin{aligned}
&\mathcal{L} = \frac{1}{2}m_2\dot{l}^2 + \frac{1}{4}m_1\dot{l}^2 - (C - m_2gl) \\
&\Rightarrow m_2\ddot{l} + \frac{1}{2}m_1\ddot{l} = m_2g \\
&\ddot{l} = \frac{m_2g}{m_2 + \frac{1}{2}m_1}
\end{aligned}
$$

6.7:With Hamiltonian

$$
\begin{aligned}
&\mathcal{L} = \frac{1}{2}m_2\dot{l}^2 + \frac{1}{4}m_1\dot{l}^2 + m_2gl \\
&p_l = \frac{\partial L}{\partial \dot{q}} = m_2\dot{l} + \frac{1}{2}m_2\dot{l} \Rightarrow \dot{l} = \frac{1}{m_2 + \frac{1}{2}m_1}p_l \\
&H = \frac{1}{m_2 + \frac{1}{2}m_1}p_l^2 - \frac{1}{2m_2 + m_1}p_l^2 - m_2gl \\
&H = \frac{1}{2m_2 + m_1}p_l^2 - m_2gl \\
&\dot{l} = \frac{\partial H}{\partial p} = \frac{1}{m_2 + \frac{1}{2}m_1}p \\
&\dot{p} = -\frac{\partial H}{\partial q} = m_2g \\
&\ddot{l} = \frac{1}{m_2 + \frac{1}{2}m_1}m_2g
\end{aligned}
$$

6.10:With Lagrangian

$$
\begin{aligned}
&q = \theta \\
&v = (R-r)\dot{\theta}; \omega r = (R-r)\dot{\theta} \\
&L = \frac{1}{2}mv^2 + \frac{1}{2}\frac{1}{2}mr^2\omega^2 - (C - mgcos\theta(R-r)) \\
&= \frac{1}{2}m(R-r)^2\dot{\theta}^2 + \frac{1}{4}m(R-r)^2\dot{\theta}^2 + mgcos\theta(R-r) \\
&= \frac{3}{4}m(R-r)\dot{\theta}^2 + mgcos\theta(R-r) \\
&\Rightarrow \frac{3}{2}(R-r)\ddot{\theta} = -gsin\theta; sin\theta \sim \theta \\
&\frac{3}{2}(R-r)\ddot{\theta} + g\theta = 0 \\
&\theta = Acos(\sqrt{\frac{2g}{3(R-r)}}t + \phi) \\
&T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{3(R-r)}{2g}}
\end{aligned}
$$

6.10:With Hamiltonian

$$
\begin{aligned}
&p = \frac{\partial L}{\partial \dot{q}} = \frac{3}{2}m(R-r)^2\dot{\theta} \Rightarrow \dot{\theta} = \frac{2}{3m(R-r)^2}p \\
&H = p\dot{\theta} - L = p\dot{\theta} - \frac{3}{4}(R-r)^2\dot{\theta}^2 - mgcos\theta(R-r) \\
&= \frac{1}{3m(R-r)^2}p^2 - mgcos\theta(R-r) \\
&\dot{p} = -\frac{\partial H}{\partial q} = mgsin\theta(R-r) \\
&\ddot{\theta} = \frac{2}{3}gsin\theta
\end{aligned}
$$